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INFO 240

17 December 2025

### Is Your Lifestyle Affecting Your Sleep

In part 3 of our project as a group we decided to test how a different lifestyle affects your sleep. As both college students and working adults we know how difficult it is to get through the day especially during the cold winter months. As a way to see if our lifestyle and daily habits affect our quality of life we decided to conduct our research on the correlation of lifestyle & sleep. In order to conduct this research we created a survey with 23 questions ranging on the amount of hours you sleep a night, to if you are on your phone before bed, your eating habits, as well as your recreational activities. To have a successful sample size, we collected approximately 60 surveys. From college students, friends, and family members to have a wide pool of data, thus eliminating age restriction from our survey. For our survey we used multiple choice questions for basic yes/no questions. In order to collect more in-depth data we used open ended questions as a way for those answering the survey to be able to give their honest opinion. Moreover, to avoid any inconclusive results we asked for each person who was answering the survey to be very specific with their answers. For every open-ended question we wrote an example of how to answer the question. For example, one of our questions was, “How much caffeine would you say that you consume daily? (please be specific with your answer for ex : I consume 8oz daily)”. In doing so, this allowed us to collect accurate data. We knew if we did not specify on how to

answer the questions we would get answers in ranges such as “10-12 oz daily”. Having an example of how to answer each open-ended question as well as specifying how to answer the question, help us avoid any inconclusive data.

## **Data Summary**

60 students' responses were used to examine how lifestyle relates to sleep. On average, students were sleeping 6.52 hours a night with a median of 7 hours. This means that most students sleep a little below the recommended 7 hours. Students spent an average of 81.92 minutes using technology before sleep but it had high variability with a standard deviation of 54.68. Students also reported exercising about 2.04 days per week on average, while experiencing stress about 3.68 days per week and setting aside time to relax about 1.24 days per week. The probability analysis shows that 46.67% of students sleep more than 7 hours per night. When we tested the conditional probability, it shows that 58% of students sleep at least 7 hours given that they consume caffeine. Using a normal distribution, approximately 66.8% of students sleep less than 7 hours a night. But when using discrete probability, we saw that only 5% of students use technology for more than two hours before sleeping.

In the correlation analysis, it showed very weak relationships between sleep and most of the variables like technology use, caffeine intake, and workout frequency. Stress showed the strongest relationship with sleep even though it was still weak and negative. So it's safe to say that higher stress may be associated with slightly less sleep. The simple linear regression using caffeine intake to predict sleep gave an  $R^2$  of 0.0013, meaning that caffeine intake doesn't

explain the difference in sleep hours. Overall, the data suggest that no single lifestyle factor strongly predicts sleep duration except for stress which even then only plays a minor role.

## **Analysis & Results**

The purpose of this project was to identify the different variables in lifestyles that may be affecting sleep patterns, and find any trends within the data collected around the topic. In considering what to analyze, it is important to acknowledge that there are multiple things that could factor into whether people may or may not be sleeping through the night. Everyone has their own habits and routines that others may not have, which changes a lot of the responses and adds multiple layers. Some of the questions that we ended up asking were their age, caffeine intake, employment status, sleep schedule, and more. These questions gave us enough feedback to portray what may affect sleep patterns the most and the different effects of these habits.

Based on the responses we got, it was obvious that there was a wide variation in what had the biggest impact on sleep between all of the participants. For example, some people had no caffeine intake, which we might associate with longer hours sleeping, but they still only got four hours of sleep. The highest number of hours someone sleeps is 9 hours, the shortest amount of hours someone sleeps is 4. We can assume that people that have different sleep patterns have gotten used to their sleep schedule and the different habits and routines that they have just worked for them. There were multiple sections of data we collected that did seem to have a correlation to the amount of sleep the participant got. These were scheduling issues such as if you are working and when you find the time to eat, and if you are dealing with stress. 53 out of 60 people said that stress affects their sleep schedules. That is a very wide range of people saying yes. Seeing the relationship between stress and amount of time sleeping was interesting because

that is not something that people have control of on a day-to-day basis. The participants didn't have high numbers on the question of being under stress throughout the week. The highest amount of caffeine a participant had was 30 ounces in the day. That seems to be a high number and they get 6 hours of sleep which is pretty good.

### **Limitations**

Within this research we had multiple questions in our survey in hopes of really getting to understand why people may be struggling with their sleep patterns. Some of the questions we asked were more helpful than others. In our results, you can see that some of the questions weren't really as strong as we thought they would be. The questions that helped us truly get a good understanding of what affects sleep was comparing the hours of sleep people normally get to their caffeine intake and seeing how stress affects their sleep. These prompts were essential to this survey considering they gave us a good view on how the different responses would help us find what could be the issue to people struggling with sleep. The questions that had significant limitations were what time of day people were consuming their caffeine, how often people work out, and especially if people use their phones before sleep considering everyone responded yes, indicating no correlation one way or the other. These questions didn't really help us find good results, as some of them weren't worded the right way, which affects the question overall.

This research was very interesting from our perspective because you can see how thought out the research project has to be. Everything has to be well coordinated or something will go wrong. The survey was what we started with, and when we first put it together we felt it was a strong survey. Now that it is completed, looking at the responses to see what actually worked, the different responses collected, and how everything came together to portray an overall picture is

interesting. Next time, it is very crucial that we plan to use better and precise wording when it comes to the survey. Seeing that the survey is the main part of the research it is very important to have the participants understand the questions better.

## **Conclusion**

In this project, our group examined whether or not there was a correlation between lifestyle factors and sleep patterns using statistical methods. The results showed that while lifestyle habits do have an influence on sleep, there is not one single factor that has a stronger influence than another. This indicates that sleep patterns are catered to each individual based on a combination of factors and how they respond to them rather than a conclusion that could be drawn for the larger population. Despite this, we could still see several trends within the data collected, for example that insufficient sleep is common amongst the students surveyed and that stress has affected the majority of the participants. If we were to conduct the survey again, we would put more focus on the clarity of the questions and gain more data to see if a correlation appears from a larger audience. Overall, the findings highlight the complex relationship between lifestyle and sleep patterns, and emphasize the importance of maintaining a balanced routine to improve sleep quality.